STORAGE DEVICES

- Floppy disk
- Hard Disk
- Compact Disc(CD)
- Flash Disk

Storage devices

- Storage devices are used to store data permanently.
- Data is stored on them for future use.
- We have different storage media:
  - Floppy disk
  - Hard Disk
  - Compact Disc(CD)
  - Flash Disk
- Each of these devices have a device that **read** data from them or **write** data onto them. This device is called **drive.**

Hard Disk

- Hard disk is the data center of PC.
- It is where all your programs and data are stored.
- A hard disk drive contains rigid, disk-shaped platters.
- The platters are coated by magnetic substance on both sides.
- Data is written to hard disk by magnetizing and demagnetizing the surface of the platters.
- Data is stored on both sides of the platter.

Types of Hard Disks

- ☑ Based on the **connectors** used to connect hard disks to motherboard, there are three types of hard disks:
  - ☑ **SCSI**
    (Small Computer System Interface)
  - ☑ **SATA**
    (Serial ATA(*Advanced Technology Attachment*)).
  - ☑ **IDE**
    (Integrated Drive Electronics)or **ATA** or **PATA**(Parallel ATA)

IDE

- ☑ It is also called **ATA**.
- ☑ In IDE, the interface electronics or controller is built into the drive itself; why it is called Integrated Drive Electronics.
- ☑ ***IDE*** is used to connect not only hard disk, but also CD-ROM drives, tape drives, and high capacity super Disk floppy drives.

Connector:
- **Parallel Cable 40 or 80 wires**

Max devices: 2
- **master**
- **slave**

Bandwidth:
- 16 MB/s originally;
- later 33, 66, 100 and 133 MB/s

Cable Select

- A drive setting called *cable select* .
- A drive set to "cable select" automatically configures itself as master or slave, according to its **position** on the cable.
- Cable select is controlled by pin **28**.
  - The host adapter grounds this pin;
  - if a device sees that the pin is grounded, it becomes the **master** device;
  - if it sees that pin 28 is **open**, the device becomes the **slave** device.

Cable Select

- This setting is usually chosen by placing a jumper on the "cable select" position, usually marked *CS*, rather than on the "master" or "slave" position.
- **Note**
  - pin 28 is only used by the drive to determine its position on the cable and thus whether it's the "master" or "slave";
  - it isn't in any way a "drive select" pin used when the host is issuing commands so if a drive is jumpered to "master" or "slave" its position on the cable doesn't have to match the pin 28 meaning of "master" and "slave".

2. SATA(Serial Advanced Technology Attachment)

- ☑ To get **faster** speed (and if your hardware supports it (hard drive and system board), you can use SATA, or Serial ATA, which will give you much faster speeds for hard drive access).
- ☑ SATA uses special cables that are NOT compatible with standard ATA hard drives.

2. SATA(Serial Advanced Technology Attachment)

- ☑ The main advantages over the older parallel ATA interface are
  - ☑ **faster data transfer**
  - ☑ **ability to remove or add devices while operating (hot swapping)**
  - ☑ **thinner cables that let air cooling work more efficiently,**
  - ☑ **more reliable operation with tighter data integrity checks.**
- ☑ SATA adapters and devices communicate over a high-speed serial cable.

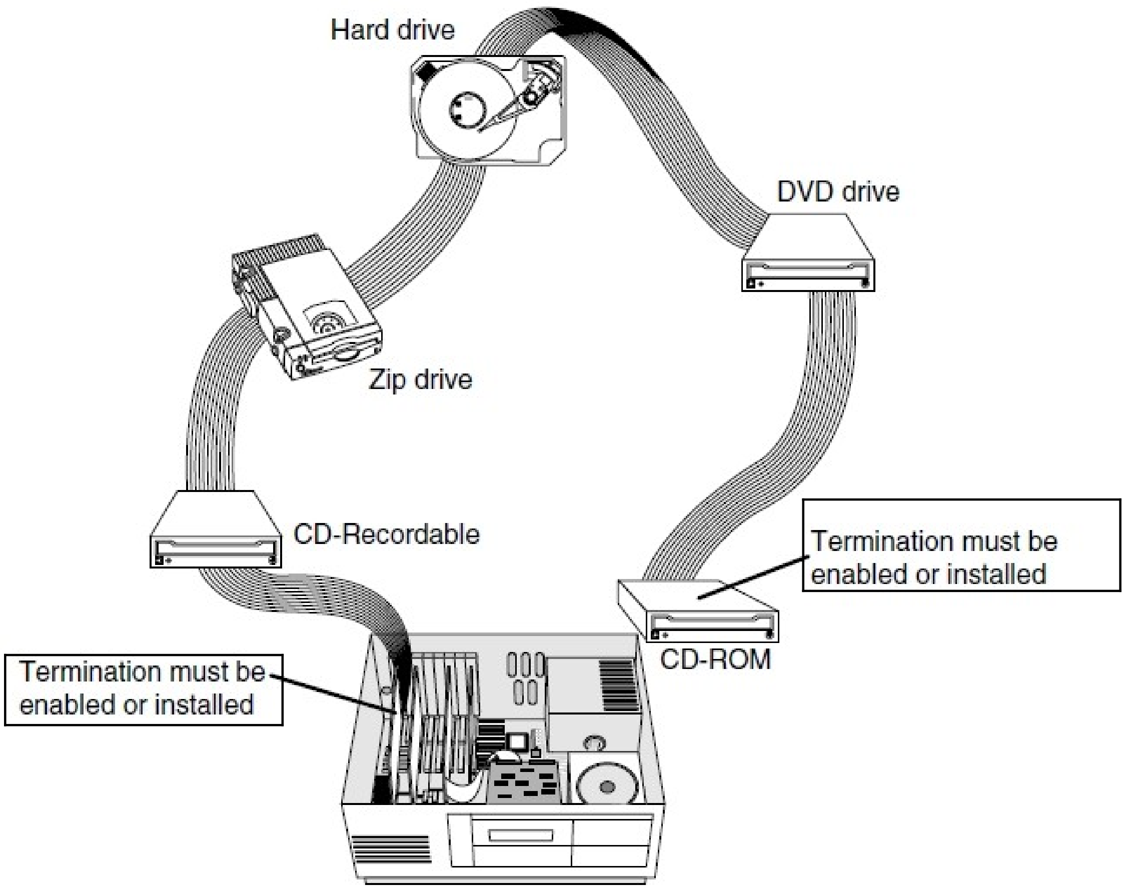

ATA
SATA

3) SCSI

- ☑ The Small Computer System Interface (SCSI, pronounced as scuzzy) standard.
- ☑ It is a technology which allows connecting different hardware devices such as hard disk drives, CD-ROM drives, tape streamers and scanners to a single adapter.
- ☑ Most systems support up to 4 host adapters, each with up to 15 devices. SCSI is a high end technology.

  - **It is a fast interface generally suited to high performance machines.**
  - **It is especially used in high end PC's such as network servers, workstations, or anywhere where storage performance is needed.**

SCSI…



Hard drive
DVD drive
Zip drive
CD-Recordable
Termination must be enabled or installed
Termination must be enabled or installed
CD-ROM

Troubleshooting Tips for HDD Problems

- ☑ Computer will not boot, and no error message appears on the screen.
- ☑ The screen remains blank when you power up the system with a blinking cursor
- ☑ The system does not recognize the drive.
- ☑ The computer hard disk is busy
- ☑ The system error message, "Drive not ready" , HDD controller failure", no OS appears.
- ☑ The drive doesn't format to full capacity.
- ☑ The Dos message "Disk boot failure", "Non system disk" or "system halted" appears.

Solution for Hard Drive

- Verify compatibility.
- Check the drive properly identified by BIOS.
- Does the hard drive spin up?
- Check the data cable
  - is bad or not
  - properly seated on either the drive or the Motherboard.
- Verify the jumper configuration.
- Check the power cable for HD.
- Take action for mechanical failure.
- Does the drive make little clicking noises?
- Check all connectors and cables.
- Check for viruses
- check the drive in other system.
- Reconfigure the hard disk
- Check the IDE controllers
- Replacement

Troubleshooting CD-ROM

- The most common causes of problems are scratches, dirt and contamination
- It is possible to clean the bottom surface of the CD with soft cloth.
  - The best technique is to wipe the disk in radial fashion, using strokes that start from the centre of the disk and emanate towards the outer edge.
  - This way the scratches will be perpendicular to the tracks rather than parallel to them minimizing the interference they might cause.
- Read errors can also occur when dust accumulates on the read lens of the drive.
- clean their drive by using canned air or standard cleaners.

If CD not recognized at boot

Solutions

- ☑ First make sure the cable is connected properly
- ☑ Check the master-slave jumpers, if both drives are on the same cable
- ☑ Check device manager for status of the drive
- ☑ If "Not ready reading drive"
  - ☑ Either you selected the CD too quickly after loading and the OS needs to catch up or the CD is dirty, clean with dry cloth.
- ☑ Use a soft clean dry cloth, never use a wet solution.

Thank You

End of the Chapter